Summary

- Ordinary Q-learning does not work well in the multi-agent setting because the environment becomes nonstationary as agents learn new policies.
- A nonstationary environment means that the expected value of rewards changes over time.
- In order to handle this nonstationarity, the Q function needs to have access to the joint-action space of other agents, but this joint-action space scales exponentially with the number of agents, which becomes intractable for most practical problems.
- Neighborhood Q-learning can mitigate the exponential scaling by only computing over the joint-action space of the immediate neighbors of a given agent, but even this can be too large if the number of neighbors is large.
- Mean field Q-learning (MF-Q) scales linearly with the number of agents because we only compute a mean action rather than a full joint-action space.